

Don't guess where to open. Ask the map.

An AI-assisted WebGIS for retail and F&B
site selection around Jakarta transit.

Pick the wrong spot and it is your capital that burns.

The blocks around stations are the busiest trading ground in Jakarta, and people still pick a location on instinct.

Nobody counted how busy it is

Everyone knows the blocks around a station are busy. Nobody knows which blocks.

The competition is invisible

Opening a coffee shop on a street already full of them is a way to lose money.

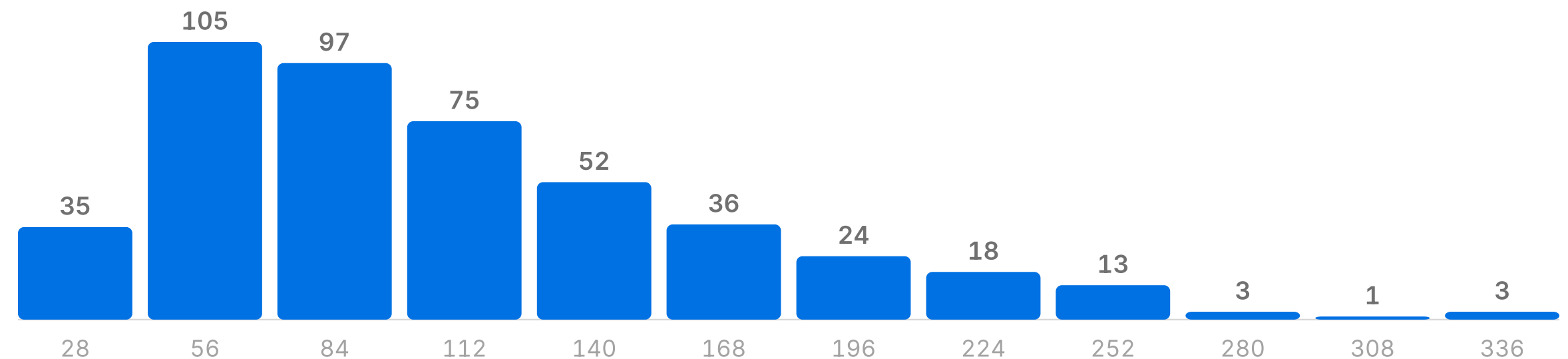
The space is never counted

A good area with nothing to rent is not an opportunity.

Busy is not spread evenly.

Each bar: how many cells hold that many businesses within walking range. Most are quiet, a few are dense.

Cells by how many businesses stand within walking range. Each column is a band, up to that many businesses, and the figure above it is how many cells fall in the band.



Find an area that is busy, still thin on your kind of rival, and has somewhere to rent.



Demand

The trade already standing around the cell, other than your kind. If a block supports dozens of businesses, people plainly come past.



Competition

Rivals of your kind within walking range, counted from two surveys. The fuller one speaks for a cell, and the two are never added.



Space

Premises actually on the market. A requirement rather than a bonus: a good area with nothing to take is not an opportunity.



Transit access

The multiplier, and what decides it most. How many stops a cell reaches within 800 m, with rail worth more than a bus.

One score per cell, per business type.

Every cell gets a label: underserved, competitive, or saturated. A ranking comes back with its reasons attached, and a place can be taken apart term by term.

Underserved

Competitive

Saturated

Busy, little space

Every cell is counted on its own.

The businesses already standing, the transit stops, and the premises on the market.
Counted one by one in each cell, not inferred from a city average.

562

cells scored
hexagonal cells, 800 m walk

1,105

transit stops mapped
MRT, KRL, LRT, TransJakarta

24,630

businesses mapped
MAPID catalogue, 13 business types, five DKI cities

13

business types scored
food, everyday retail, and services

5,711

competitor points from OpenStreetMap
9 business types, all of Jakarta

3,547

premises on the market
asking prices for sale, 284 cells with a median

709

field notes that landed in a cell
Struk Go, Menu Go, Properti Go and community notes, 191 cells

3,047

readable opening timetables
of 19,421 businesses counted, 3,129 publishing hours

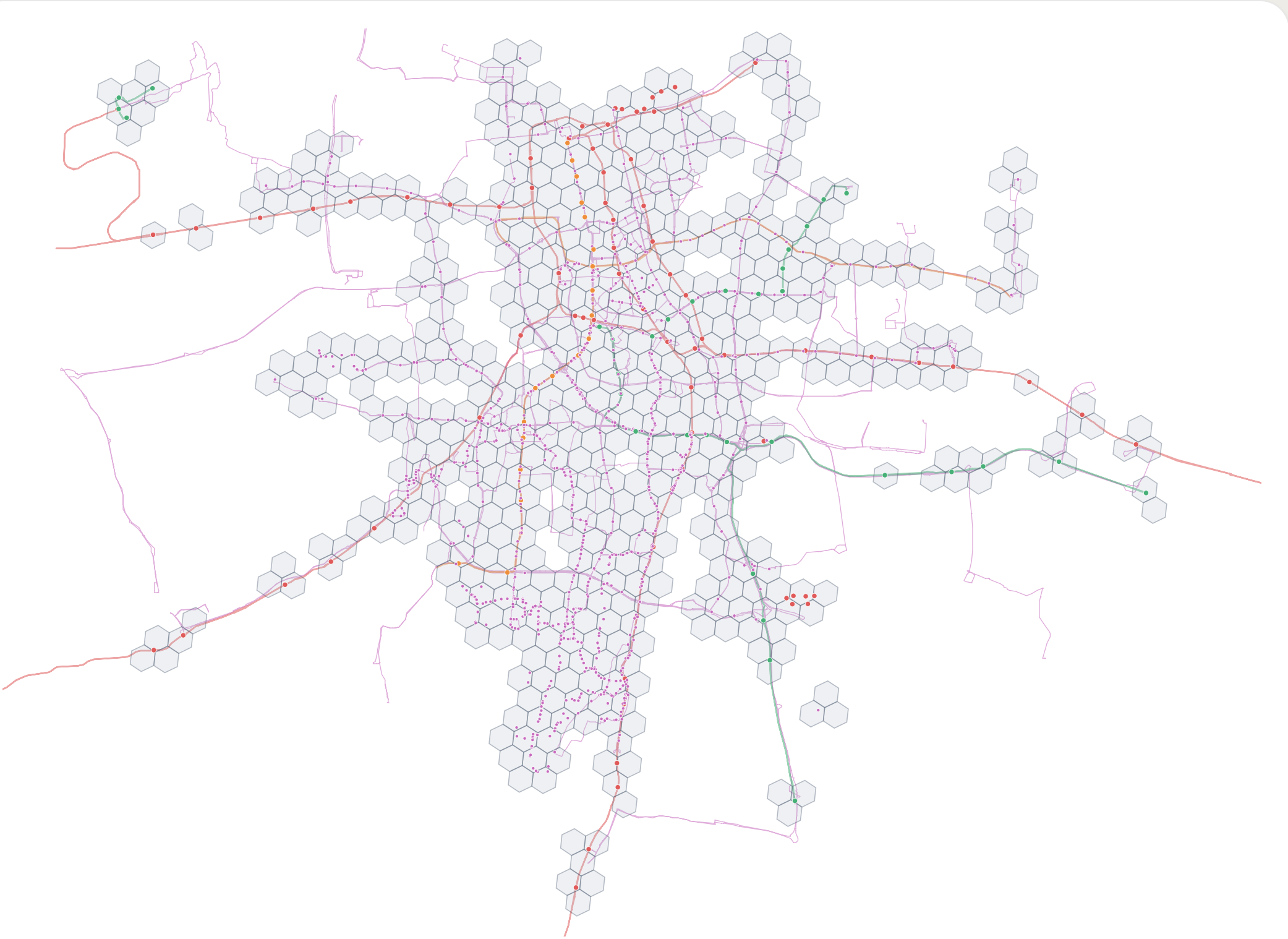
Cells one walk wide, one hexagon per cell.

Jakarta is covered with an H3 grid at resolution 8, about 537 m to a side. Only the **562 cells** with a transit stop within 800 m are scored, and the score reads everything within that walk of the cell centre.

Not one catchment per stop, on purpose. TransJakarta stops sit a few hundred metres apart while the walking range is 800 m, so per-stop catchments overlap and count the same shoppers over and over. On a grid each cell is counted once, and transit access becomes a property of the cell: a place served by both the MRT and TransJakarta really is worth more.

● MRT	20
● KRL	76
● LRT	33
● TransJakarta	976

1,105 transit nodes and the corridor geometry, from OpenStreetMap



Every hexagon at its real place. Corridors and stops in their mode's colour, the grid drawn by its edges as it is before a question is asked.

Three kinds of data, kept apart.

Two of them claim completeness for the city they cover, so a zero from them is a finding. The third is streets somebody walked, and an empty cell there only means nobody has been.

OPENSTREETMAP · ODBL

Transit, competitors, hours

1,105 transit nodes in four modes and the corridor geometry. 5,711 competitor points in 9 business types. 3,047 readable opening timetables, read as doors open per hour.

MAPID PREMIUM CATALOGUE

Businesses and premises

24,630 business points in all 13 types across the five DKI cities, every one carrying its name. 3,547 premises on the market, as asking prices to buy. Coverage is decided per administrative city.

MAPID APPS FIELD MISSIONS

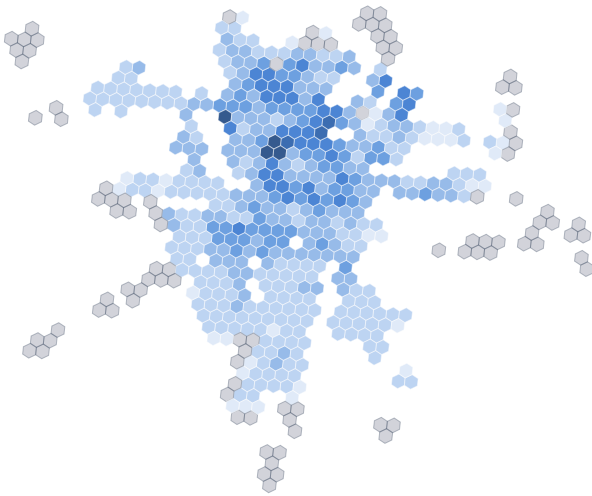
Struk Go, Menu Go, Properti Go

1,027 records read, 709 landed in 191 catchments: 98 receipts, 61 eateries, 54 premises and 496 community notes. Evidence beside the score, never a term in it.

A grid of 562 cells: 462 sit in a surveyed city, 100 do not.

A cell whose city the catalogue has never read is drawn hatched and left unscored. It is never interpolated, never rounded to zero, and never quietly dropped from a ranking. The five cities: Jakarta Selatan 178, Jakarta Timur 105, Jakarta Barat 74, Jakarta Pusat 58, Jakarta Utara 47.

● cells with data 462 ▨ no data yet 100



How this score is made

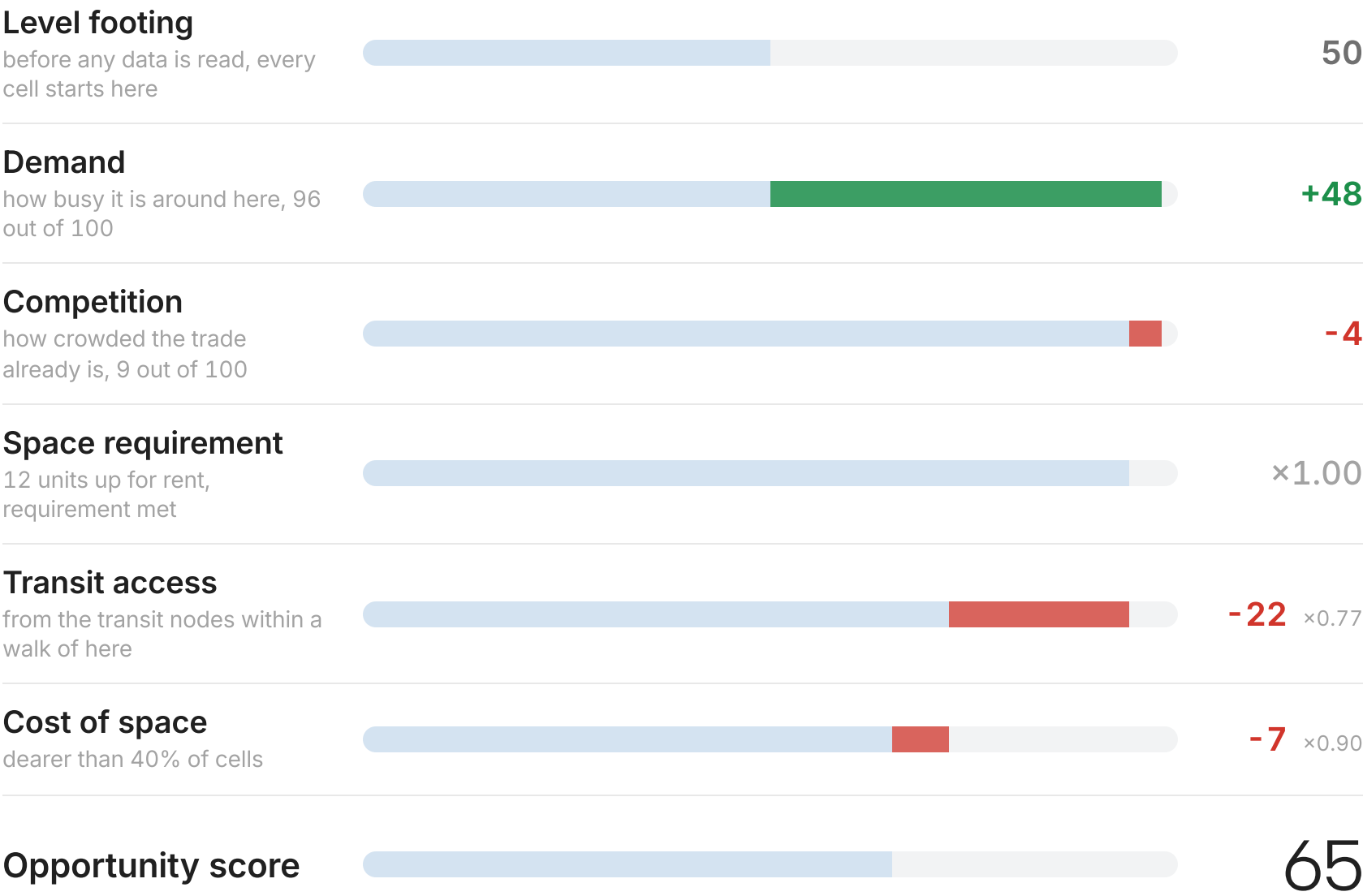
$$\text{Gap} = (w_d \cdot \text{demand} - w_s \cdot \text{supply}) / (w_d + w_s)$$
$$\text{Score} = \text{clamp}(\text{Gap} + 0.5) \times \text{gate} \times \text{access} \times \text{cost}$$

Demand	the trade around the cell other than your kind, as a share of the busiest cell on the grid
Competition	rivals of your kind, as a share of the densest street for them. Weighted 0.5 and 0.5 by default, and the reader can move both
Space gate	×1 where premises are on the market, ×0.15 where none are. Pushed to the bottom of the ranking, not struck off it
Transit access	×0.6 with no stop in range, up to ×1.0. MRT 1, KRL 0.9, LRT 0.6, TransJakarta 0.45 per stop
Cost of space	×1 down to ×0.75, a rank among the catchments that carry a price. Never above 1, so an unpriced place is never handed a price to be judged on

The score above is built in order. Every step can be traced back to its data. Every term is counted, none is generated.

RAWA SELATAN · COFFEE SHOP · 800 M

What decides it most: mass transit



Of this cell's 65 points, 14 come from its transit access. The score above is built in order. Every step can be traced back to its data.

The answer is computed, and then it is said.

Say what you want to open and the whole city recolours for that business. No formula to fill in and no jargon to memorise.

THE QUESTION

Asked in plain language

Indonesian or English, typed or tapped. The turns before it travel with it, so a follow-up that only points, why that one, is read in context.

UNDERSTANDING

The model picks an operation

Through function calling it names a shape, a measure and the arguments, never a value. A question outside the data comes back as not understood. Without a key, or after 90 seconds, the rule parser takes over.

COMPUTING

The scoring engine on the grid

Every figure a reader sees, score, demand, rivals, price, standing, is computed on the data in one go. No figure is ever streamed a digit at a time.

WRITING

The model writes the sentence

From a fact sheet of the figures this turn produced. A reply carrying a figure or a place not on the sheet is thrown away whole, and the composed sentence stands in.

THE SCREEN

The map repaints

The answer names places, the whole city recolours for the business type, and every reply says which path read it, the model or the rules.

No figure a reader sees came from the model.

The model decides what to look up and how to say it. The arithmetic is the engine's alone, and the fence checks every sentence against the figures it was handed. Five shapes of question, rank, flag the saturated, compare, coverage and explain, across 11 measures, chosen separately.

WHAT STREAMS WHILE THE READER WAITS: THE STAGE, NEVER A FIGURE

reading

One moment, I am reading your question

retrying

Still no answer, trying another way

choosing

I have it, working out what to look up

computing

Now counting it up from my notes

writing

I have the figures, putting the answer together

Reported from where the work actually is, never on a timer and never as a percentage. Up to 90 seconds for understanding across a chain of 5 free models, 60 for the written reply.



Meet Tapak, your guide.

Tapak reads the question, works out what to look up, then answers with the reasons attached. The model writes the sentence. It does not write the figures: every one of those is computed by the scoring engine on the data, and a figure that is not among them is thrown out before it reaches the screen. Where a catchment has never been surveyed, Tapak says so rather than filling the gap.

The waiting state is the figure.

A spinner says the machine is busy. Tapak pacing says somebody is looking it up, and it is the same figure that answers.

Small talk is allowed, and fenced.

Hello, what SpotOn is, and general business talk. Any digit in such a reply and it is thrown away, because nothing was computed behind it.

The map never moves on a casual turn.

No items, no highlight, no change of business type. Only an answer computed from the data can repaint the city.

Good morning. I'm Tapak. I've been round 562 cells near the MRT, KRL, LRT and TransJakarta corridors. What are you thinking of opening?

Coffee shop

Drinks & dessert

Bakery

Warung & rice shop

Fast food outlet

Noodles & meatballs

Seafood restaurant

Foreign cuisine restaurant

Minimarket

Grocery store

Laundry

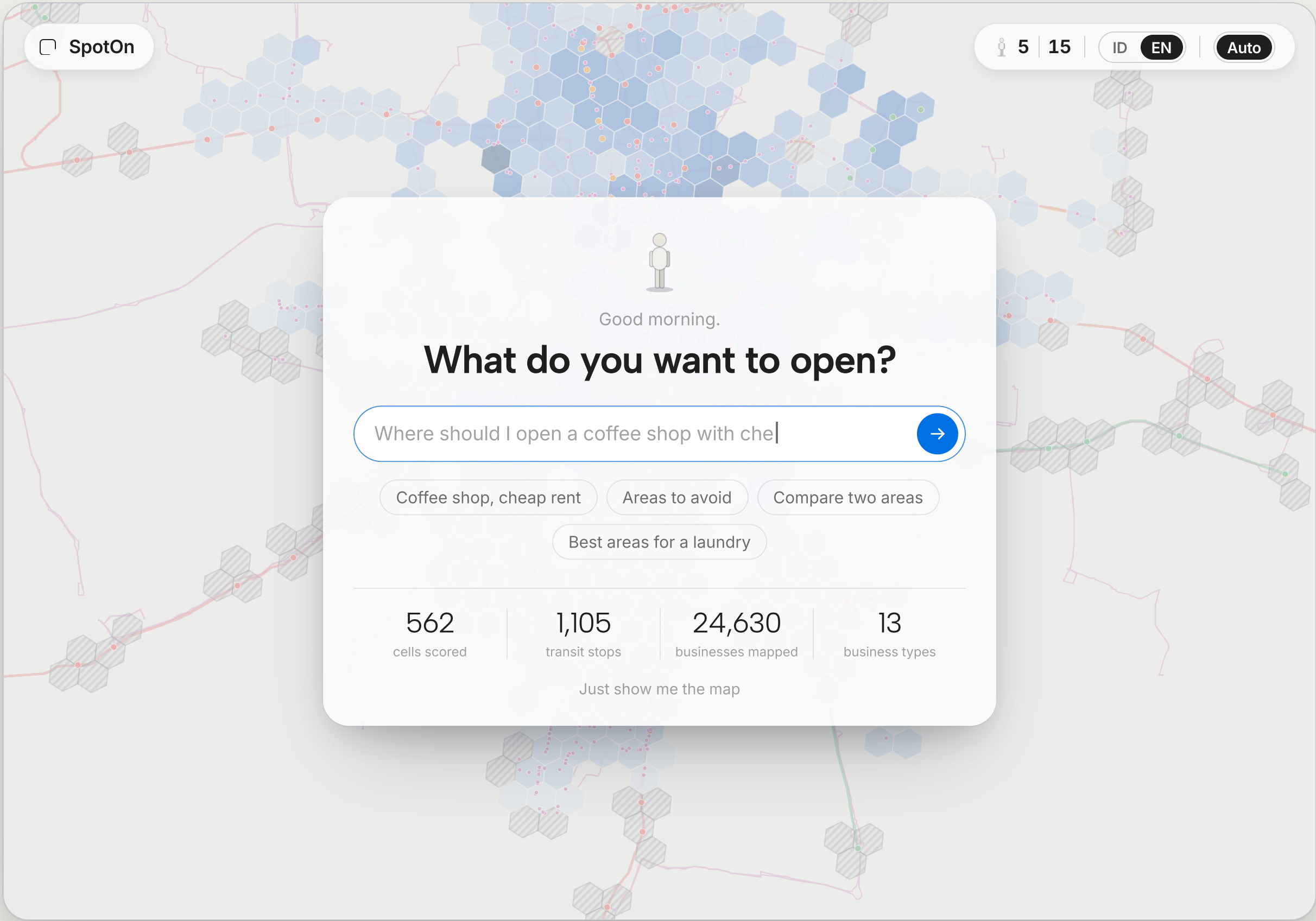
Repair shop

Pharmacy

Coffee shop

Now counting it up from my notes ...

The greeting follows the clock on the reader's own device and knows nothing else: not the weather, not whether the street is busy. Both surfaces that say hello read one table, so they cannot disagree about the hour.



Act one. One question, where the eye already is.

Nothing to fill in.

No rail of thirteen buttons, no sliders, no table. A greeting by the hour, the one question the reader is here to answer, and a field that takes anything.

The examples teach the shape.

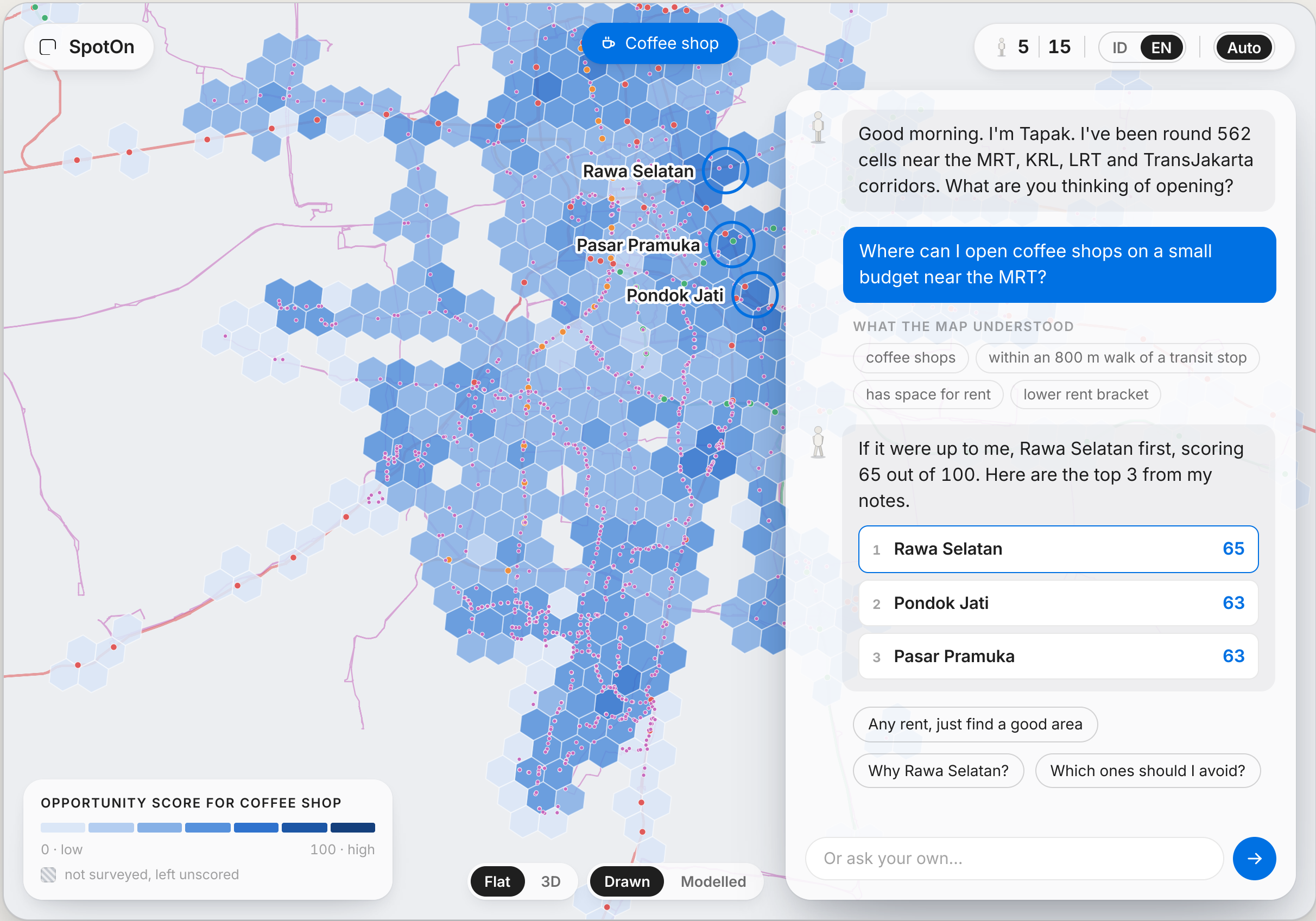
They type themselves into the field and clear again, and the same four sit under it as buttons, so nobody has to type to get started.

The evidence is on the card.

Four counts at the foot, read from the grid file's own metadata. Rebuild the grid and they follow.

The map is free.

Panning, colouring, switching business type, moving the range: none of it is metered. Behind the box the city is already painted by the trade around each cell.



Act two. Ask, and the whole city recolours.

The box flies right and becomes the thread.

The map takes over. Every place the answer names is marked on it, and tapping one opens its card.

What the map understood comes first.

Before answering, the map shows what it understood. If it picked something up wrong, you see it immediately.

Every figure in the list is the engine's.

Rawa Selatan leads at 65 out of 100, computed on the grid just now, with lower rent bracket as the only narrowing.

The follow-ups are real follow-ups.

Why that one, any rent, which ones to avoid. Each is read against the thread, and the reply says which path read it.



Rawa Selatan

65

● Underserved
 the highest of all areas


 Ask Tapak about this area



HOW THIS SCORE IS MADE

Coffee shop

Opportunity score the highest of all areas	65	Busyness higher than 99% of areas	96
How crowded the trade is higher than 42% of areas	9	Other businesses nearby	273
Competitors of this kind	5	Transit access higher than 56% of areas	42
Units on the market	12		



WHAT THIS AREA REACHES

4

 transit nodes within walking range

4 TransJakarta stops. The area's transit access is fair.



COST OF SPACE

Rp 40.2m

 per m² of land · 12 units on the market

Dearer than 40% of the cells with a readable price. That price takes 7 points off this cell.



WHEN THIS AREA IS OPEN

4 of 54 publish readable hours

Only 4 of the 54 businesses on record within 800 m publish opening hours. Too few to show how the day goes.



FIELD NOTES

1 record in this area. These are notes from people who went and stood there, not a census. An area with nothing recorded is one nobody has walked yet.

Tap a cell, and it is taken apart.

An index says where it sits.

Each "higher than" sets this area against every other area that has that figure, for the same business type and walking range. A count is its own comparator, so the counts say nothing extra.

The price is a price to buy.

This is an asking price to buy, not a rent. The MAPID catalogue carries no rental listings for Jakarta at all, so there is no monthly rent to show.

The curve counts doors, not people.

For each hour, how many businesses within walking range publish a timetable that says they are open. The denominator travels with it, because fewer than one business in six publishes hours at all.

Field notes are recorded, not counted.

Receipts, what a meal costs, how busy the place looked, space being offered. Photographed, so the street can be seen first, and kept out of the score.

Figures for Rawa Selatan, coffee shop, 800 m walking range, computed by the engine on the grid at build time.

Any area can be seen as a model.

Its buildings, streets and stops are read off the same basemap, cut to the walking range, and can be turned by hand. Each building stands at the height the map records.

Read, not composed.

The basemap's own vector tiles, cut to a disc of the walking range around the point the range is measured from. A street the tile does not draw is not drawn.

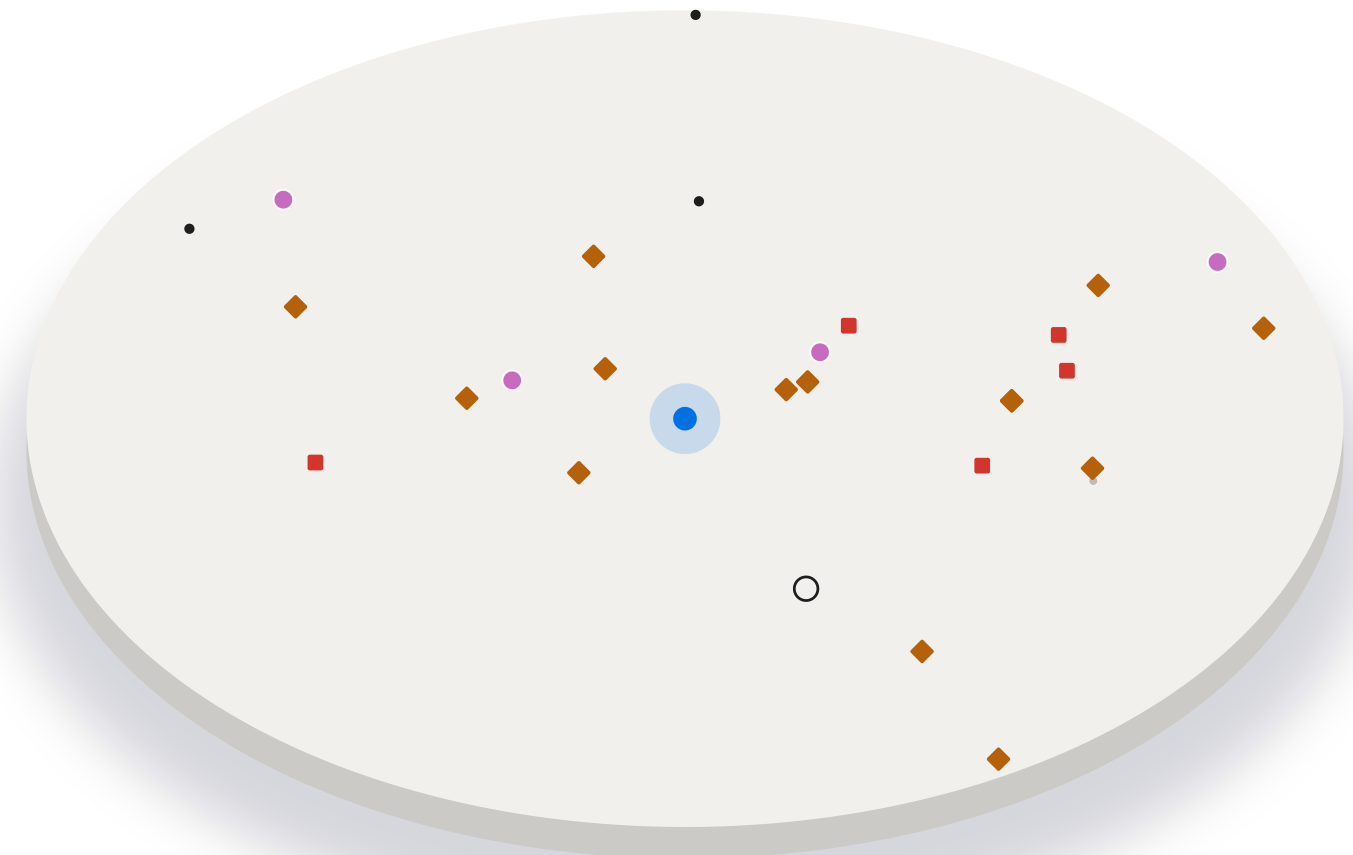
Doors light by the hour.

One mark per business with readable hours, lit when its timetable says it is open in the hour on the slider. Where the doors were not counted, none is drawn and the view says so.

A miniature, not a view.

A disc on a base with a rim, turned by hand, with the sun staying where it is in the world. The map itself can be looked at the same way, drawn or modelled.

Rawa Selatan



- the point the range is measured from
- transit nodes, 4
- coffee shop rivals, 5
- ◆ units on the market, 14
- field records, 1
- doors open at 10a on a Monday, 3 of 4

Every mark at its real position within 800 m of Rawa Selatan. The buildings, streets, water and parks are read from the basemap tiles the moment an area is opened, so they are not drawn here.

What it refuses to do.

1 **Read a null as a zero.** An unsurveyed city is not an empty one. It is dropped from a ranking rather than sorted to the bottom, where it would crown the least examined streets the best opportunities.

3 **Type a figure into copy.** Every count on every page, this deck included, is read from the grid's metadata or computed from it. A number written by hand once went stale in silence.

5 **Count people.** Nobody has counted a person for this product. The activity chart counts doors open per hour from published timetables, and refuses a timetable it can only half read.

7 **Let the model write a number.** A written reply may carry only figures and places it was handed, and one that does not is thrown away whole. A rejection is logged with its reason, never silent.

The product's whole claim is that its figures do not come from anybody's memory. These are the rules that keep it true, and each of them cost something to learn.

2 **Invent a threshold.** A filter names a band, the cheapest third or the busiest, cut from the grid's own distribution at query time. There is no way to say under thirty million, because that figure would come from nobody's data.

4 **Call a sale price a rent.** The catalogue publishes no rent for Jakarta, re-checked on every fetch. So the cost of space is an asking price to buy, named as such wherever it travels, and it only ever deducts.

6 **Let a survey into the score.** Field notes ride beside the figures as evidence. Folded into the arithmetic, nobody went here would be identical to nothing happens here.

8 **Guess.** A question the data cannot answer comes back as not understood, and a sentence the rule parser reads nothing in is answered by saying what can be asked, never by the default ranking.

Six minutes, from the front page to a shortlist.

What to click, and the one sentence to say at each step. Every figure that appears on screen is the engine's, so nothing has to be rehearsed except the order.

- 1

Open the front page and scroll one block.

The street model walks through a day as the page scrolls. **Say:** busy is something you can count. The four figures above are read from the grid file, not typed.
- 3

Sign in and open the app.

One button in demo mode, no password. Tapak greets by the hour. **Say:** what do you want to open?
- 5

Ask “Why Rawa Selatan?”

Typed or tapped. The score comes back taken apart, transit first. **Say:** the model wrote this sentence, the engine wrote every figure in it.
- 7

Open the model, turn it, drag the hour.

Then switch the map to 3D and to Modelled. **Say:** a building stands at the height the map records. Nothing on this disc was composed.

- 2

Let the conversation play.

The map beside it recolours for each business type. **Say:** the questions are samples, the answers are computed by the same engine the app runs.
- 4

Tap “Coffee shop”, then “Only where the rent is cheap”.

The city recolours, the top three arrive, the chips show what was understood. **Say:** the filter is a band from the grid's own thirds, never a number the model chose.
- 6

Tap the top cell and walk the card.

Standing against the grid, the nodes it reaches, what space costs, when the doors are open, what somebody recorded there. **Say:** each index says where it sits among every area that has one.
- 8

Switch language mid-answer, then open the account page.

Indonesian and English are one dictionary in two shapes. The account page shows the two meters. **Say:** the map is never metered, only a question and an opened place are.

Have ready

A Map Service key for the MAPID basemap, or the open one stands in. A model key on OpenRouter, or the rule parser answers the plain forms and says so. An empty database address, which is demo mode. If the network is slow, the stages on screen say where the wait is.

No key is a way of running this.

Openable from a bare clone with nothing filled in. Each missing secret costs exactly the thing it unlocks and nothing else, and the interface says which mode it is in.

NO DATABASE

Demo mode

One account, one button to sign in, records in memory until the process restarts. The same tiers, the same allowances, the same meters and session tokens as with a cluster behind it.

NO MODEL KEY

The rule parser

It reads the plain forms: a business type, a measure, a named place, a why. Every figure is still the engine's, and every reply says which path read it rather than glossing over it.

NO MAP SERVICE KEY

An open basemap

OpenStreetMap data with CARTO cartography stands in, in the reader's theme. The MAPID basemap is drawn the moment a key is present, and is what the final product ships on.

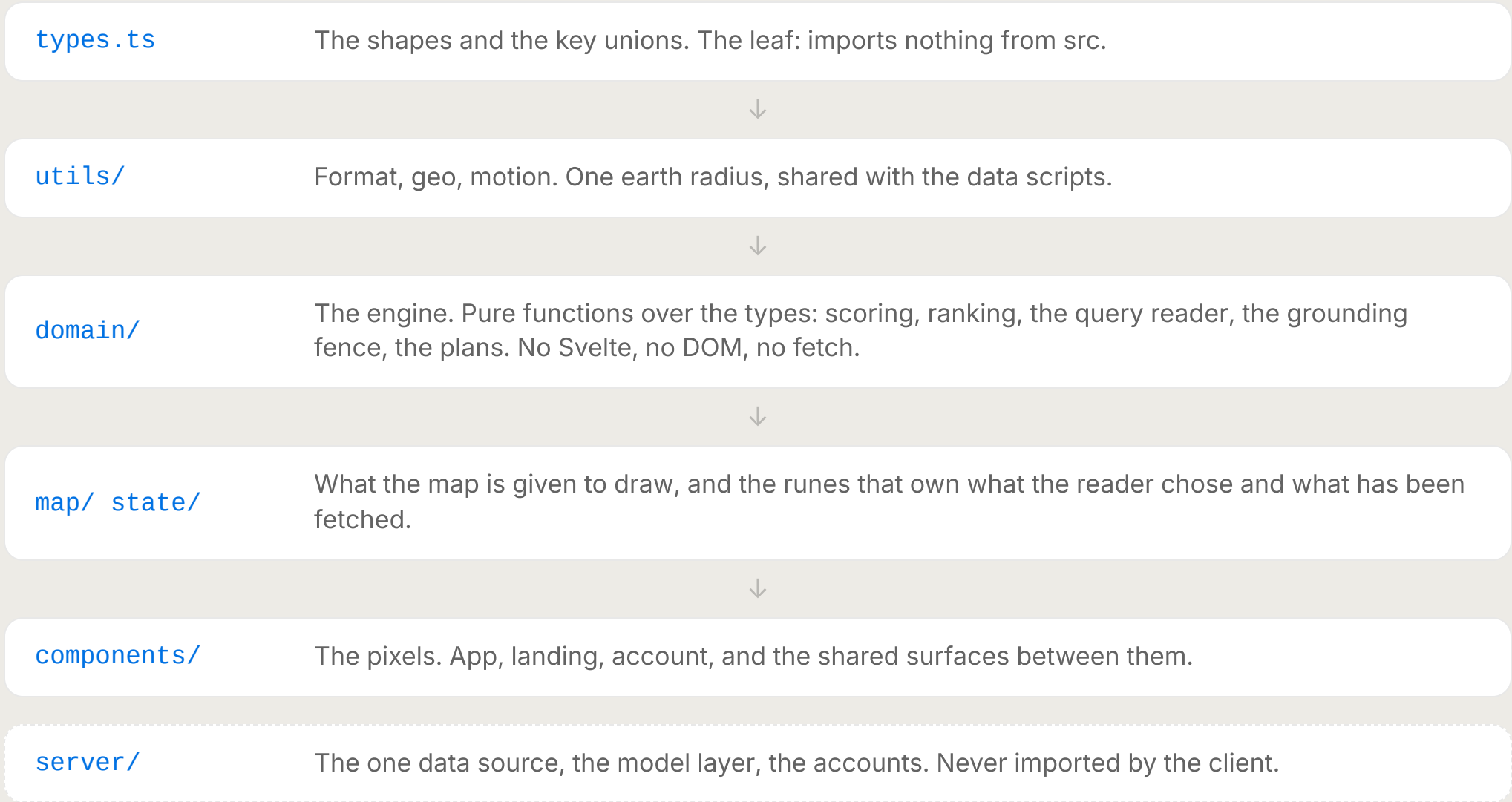
Two metered actions, and the map is never one of them.

One question to Tapak, because understanding a sentence is a call out to a shared model. One area or place opened by hand, because that is when its competitors, stations, listings and timetables are all read and drawn. Closing costs nothing and reopening what is open costs nothing. The week turns at midnight on Monday in Jakarta, and what was bought outright survives every Monday.

PLAN	QUESTIONS A WEEK	AREAS OR PLACES A WEEK	RP A MONTH
Free	5	15	0
Personal	60	300	79,000
Premier	250	1,500	249,000
Top - up, questions	50	·	49,000 once
Top - up, areas or places	·	300	39,000 once

Arrows only point downwards.

Every layer imports only from the layers beneath it, so the engine can be run on the server, in the browser and in a build script alike. This deck is built by loading the same modules and asking them.



STACK

SvelteKit, Svelte with runes	2.63.0 · 5.56.1
TypeScript, Vite	6.0.3 · 8.0.16
MapLibre GL, the map and the modelled basemap	6.3.0
three.js, the miniatures and the street model	0.185.1
H3, the grid	4.5.0
MongoDB, optional, accounts and meters	7.5.0
OpenRouter, function calling, a chain of free models	5

DATA

Built by scripts over Overpass and the MAPID catalogue and missions, in an order each file records in its own metadata under a regenerate field. Every distance is measured with one earth, the one the browser measures with.

CHECKS

Typecheck, then eleven self-tests that hold the pieces against each other, the breakdown against the engine, the two hour passes against every cell, the fence against the composed sentences, then the production build. Green on every push, deployed to Vercel from main.



A map that answers, not one that just displays.

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Bina Nusantara University

Transit and business data © OpenStreetMap contributors (ODbL). Business points, property
listings and field notes from MAPID. Built on 2026-09-10 from the grid in the repository.
Rebuild with `npm run deck`.